10.5 Infinitely Many Bounces in Finite Time

WORKSHEET

The question

STEP III 2002, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself When a particle hits a smooth wall with coefficient of restitution e, what happens to the component of velocity along the wall? And the component perpendicular to it?

STAGE 1 The angle of departure

Ask yourself The particle arrives at M_1 at angle θ to the wall OA (measured on the side away from O). At what angle φ to OA does it leave? Use the triangle OM_1N_1 to find φ in terms of θ.

STAGE 2 The pattern repeats

Ask yourself At N_1, the particle arrives at angle θ to the wall OB, measured on the side away from O. Compare this with its arrival at M_1. What happens next?

STAGE 3 Two ratios

Ask yourself By what factor ρ does the distance from O shrink from one bounce to the next? By what factor k does the speed shrink? So by what factor does the time for each leg shrink?

STAGE 4 A geometric series

Ask yourself When is $\sin(2\theta - 60^\circ) < \sin 2\theta$? Use the identity $\sin X - \sin Y = 2\cos\frac{1}{2}(X+Y)\sin\frac{1}{2}(X-Y)$.

What's really going on

When $\theta \geq 60^\circ$

At $\theta = 60^\circ$, the ratio is exactly 1: every leg takes the same time, and the particle never reaches O, although it gets ever closer. (The speed and the leg lengths both shrink by the same factor.) For $\theta > 60^\circ$, the legs take longer and longer. In every case the particle approaches O and makes infinitely many bounces; only for $\theta < 60^\circ$ does it get there in finite time. The value $e = \tan 30^\circ / \tan 60^\circ = \frac{1}{3}$ is the dividing line.

Bouncing balls and inelastic collapse

The simplest example of the same phenomenon: a ball dropped onto a floor rises to height e^2H, then e^4H, and so on. The time between bounces is proportional to the speed at each bounce, which shrinks by the factor e, so the total time is a geometric series:

with $T_0 = \sqrt{2H/g}$ the time of the first fall, the ball stops bouncing after

$$T_0 + 2eT_0 + 2e^2T_0 + \cdots = T_0\,\frac{1+e}{1-e}.$$

Infinitely many bounces, and then rest. In 1992 Sean McNamara and William Young showed that the same thing happens in simulations of many inelastic particles in a line: a group of them can undergo infinitely many collisions in a finite time, an "inelastic collapse". Simulations of sand and granular flows must include special rules to avoid it, and the phenomenon helps explain why granular gases spontaneously form clusters.

Zeno and convergence

Zeno's paradoxes, from around 450 BC, troubled philosophers for two thousand years. The mathematical resolution, that an infinite sum can have a finite value, was made precise only in the nineteenth century, when Cauchy, Weierstrass and others defined convergence by means of limits (the subject of Chapter 2). The surprising feature of this question is that the process is *physically* infinite: there really are infinitely many collisions, in a finite time. Mathematicians studying billiards in polygons and the motion of colliding particles call such behaviour a *Zeno phenomenon*, and it is one of the subtleties in proving that systems of colliding hard spheres have well-defined motions for all time.

Try this next

1. **The total distance.** Show that the total distance travelled from M_1 to O is finite for every θ between 30° and 90°, and find it in terms of OM_1 and θ.

2. **Other wedges.** Repeat the question for a wedge of angle α instead of 30°. Show that $\varphi = \theta - \alpha$, and find the condition on θ for the particle to reach O in a finite time.

3. **Elastic walls.** If $e = 1$, can the particle ever reach O? (*Hint:* reflect the wedge repeatedly in its walls, so that the path becomes a straight line. How many copies of a 30° wedge fit round O?)

4. **The bouncing ball.** A ball is dropped from height 1 m with $e = 0.8$. How long does it bounce for, and how far does it travel in total? (Take $g = 9.8\ \mathrm{m\,s^{-2}}$.)

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5. Includes several problems on bouncing balls and infinite sequences of collisions.
- **Sean McNamara & William Young, "Inelastic collapse and clumping in a one-dimensional granular medium"**, *Physics of Fluids A* 4 (1992).
- **Serge Tabachnikov, *Geometry and Billiards*** (American Mathematical Society; free online). The mathematics of billiards, including the unfolding trick in the exercises.

The full guided solution is in Section 10.5:
https://beyondstep.keithmcnulty.org/ch10/zeno-wedge.html